M10e - Resonance and Phase Shift in Mechanical Oscillations

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1 Introduction

Mechanical oscillations are a fundamental class of motions that occur in many physical systems, from simple pendulums and springs to bridges, buildings, and microscopic resonators. In the idealized case of a harmonic oscillator without friction, the system oscillates indefinitely with a fixed amplitude and frequency determined solely by its inertial and restoring properties. Real systems, however, are always subject to dissipative forces that remove energy from the motion, leading to damping and a gradual decay of the amplitude. When such a system is additionally driven by an external periodic force, rich phenomena such as resonance and frequency-dependent phase shifts between drive and response arise.

In this experiment we investigate these effects using Pohl's wheel, a rotary pendulum consisting of a copper disk attached to a spiral spring. The restoring torque of the spring and the moment of inertia of the disk define the eigenfrequency of the undamped system, while an adjustable eddy-current brake introduces a controllable damping torque. An electric motor provides a periodic driving torque, allowing us to study both free (undriven) and forced oscillations within the same apparatus. Angular deflections of the wheel and of the drive are measured by magnetic angle sensors and recorded via an analog-to-digital converter.

The experiment is divided into two main tasks. In the first task, we study the *free*, damped oscillation of the rotary pendulum for several values of the damping set by the coil current of the eddy-current brake. From the time evolution of the angular deflection we determine, for each damping setting, the frequency of the damped oscillation and the damping constant. By analysing the dependence of the damped angular frequency on the damping constant, we extract the eigenfrequency of the undamped system and compare our findings with the theoretical relation between these quantities.

In the second task, we investigate the *forced* oscillation of the rotary pendulum driven by the motor. For a fixed damping, we vary the driving frequency and measure both the steady-state oscillation amplitude of the wheel and the phase shift between the drive and the response. This allows us to record the resonance curve, i.e. the frequency dependence of the amplitude, as well as the corresponding phase shift as a function of driving frequency. By fitting the theoretical expressions for a driven, damped harmonic oscillator to the measured amplitude and phase data, we determine the resonance frequency and the damping constant and compare them with the values obtained from the free oscillation.

A further aim of this work is to quantify the reliability of our results by performing a systematic uncertainty analysis. We estimate statistical uncertainties arising from the finite sampling rate and peak determination in the time series, as well as systematic contributions due to the calibration of the angle sensors and the settings of the current and voltage supplies. These uncertainties are propagated through the analysis and included in the final values reported for the eigenfrequency, resonance frequency, and damping constant.

2 Theoretical Analysis

2.1 Overview

The Pohl's wheel used in this experiment behaves as a *rotary pendulum*, i.e. a torsional harmonic oscillator consisting of a rigid body with moment of inertia J, a spiral spring generating a restoring torque proportional to the angular deflection φ, and a damping torque arising from eddy currents. The system can perform two types of motion relevant to this experiment:

1. **Free damped oscillation**, where the pendulum is displaced and allowed to oscillate without external driving torque.

2. **Forced oscillation**, where the pendulum is driven by a periodic external torque of angular frequency ω.

This section provides the theoretical foundations needed to interpret both tasks: moment of inertia, equation of motion, solution of the damped oscillator, logarithmic decrement, resonance curve, and the frequency-dependent phase shift.

2.2 Moment of Inertia and Angular Momentum

The moment of inertia of a rigid body is defined as

$$J = \sum_i m_i r_i^2, \tag{1}$$

where r_i is the distance of the mass element from the axis of rotation. For a homogeneous disk of mass m and radius R, the moment of inertia is

$$J_{\text{disk}} = \frac{1}{2} m R^2. \tag{2}$$

The angular momentum of a rotating rigid body is

$$L = J\dot{\varphi}, \tag{3}$$

and the torque is given by

$$M = \frac{dL}{dt} = J\ddot{\varphi}. \tag{4}$$

2.3 Rotary Pendulum and Torque Balance

Three torques act on the Pohl's wheel:

- **Inertial torque:**
  $$M_T = J\ddot{\varphi}.$$
- **Restoring torque** from the spiral spring:
  $$M_F = -D\varphi,$$
  where D is the directional spring constant.
- **Damping torque** due to eddy currents:
  $$M_D = -\gamma\dot{\varphi},$$
  with damping coefficient γ.

The torque balance is (manual Eq. (1)):

$$J\ddot{\varphi} + \gamma\dot{\varphi} + D\varphi = 0. \tag{5}$$

Introducing the natural angular frequency and damping constant,

$$\omega_0 = \sqrt{\frac{D}{J}}, \qquad \delta = \frac{\gamma}{2J}, \tag{6}$$

the equation of motion becomes

$$\ddot{\varphi} + 2\delta\dot{\varphi} + \omega_0^2\varphi = 0. \tag{7}$$

2.4 Solution of the Damped Oscillator

For weak damping, i.e. $\delta < \omega_0$, the solution of Eq. (7) is

$$\varphi(t) = A e^{-\delta t}\cos(\omega_d t + \alpha), \tag{8}$$

where

$$\omega_d = \sqrt{\omega_0^2 - \delta^2} \tag{9}$$

is the frequency of the damped oscillation.

2.4.1 Logarithmic Decrement and Damping Constant

Let $\varphi(t)$ and $\varphi(t + T_d)$ be two successive peak amplitudes, where the damped period is $T_d = 2\pi/\omega_d$. The logarithmic decrement is defined as

$$\Lambda = \ln\left(\frac{\varphi(t)}{\varphi(t + T_d)}\right). \tag{10}$$

Since the amplitude decays as $e^{-\delta t}$, we obtain

$$\Lambda = \delta T_d \qquad \Rightarrow \qquad \delta = \frac{\Lambda}{T_d}. \tag{11}$$

This allows experimental determination of δ directly from peak values.

2.4.2 Relation Between ω_d and δ

Rearranging Eq. (9), we obtain

$$\omega_d^2 = \omega_0^2 - \delta^2. \tag{12}$$

This relation is central to Task 1: plotting ω_d^2 vs. δ^2 yields a straight line whose intercept gives ω_0^2.

2.5 Forced Oscillation and Steady-State Response

When the pendulum is driven by a periodic external torque

$$M_{\text{ext}}(t) = M_0 \sin(\omega t), \tag{13}$$

the equation of motion becomes (manual Eq. (8)):

$$J\ddot{\varphi} + \gamma\dot{\varphi} + D\varphi = M_0 \sin(\omega t). \tag{14}$$

The steady-state solution has the form

$$\varphi(t) = A(\omega)\,\sin(\omega t + \theta), \tag{15}$$

where both the amplitude $A(\omega)$ and the phase $\theta(\omega)$ depend on the driving frequency.

2.6 Resonance Curve

Solving Eq. (14) (see manual p. 3–4), the amplitude is

$$A(\omega) = \frac{M_0/J}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(2\delta\omega\right)^2}}. \tag{16}$$

The resonance frequency is defined as the value of ω that maximizes $A(\omega)$. Setting the derivative to zero yields (manual Eq. (13)):

$$\omega_R = \sqrt{\omega_0^2 - 2\delta^2}. \tag{17}$$

For weak damping, $\delta \ll \omega_0$, this reduces to

$$\omega_R \approx \omega_0. \tag{18}$$

2.7 Frequency-Dependent Phase Shift

The phase shift between the driving motion and the pendulum is given by

$$\tan\theta(\omega) = \frac{2\delta\omega}{\omega_0^2 - \omega^2}. \tag{19}$$

From Eq. (19), two regimes follow:

$$\theta(\omega) = -\arctan\left(\frac{2\delta\omega}{\omega_0^2 - \omega^2}\right), \qquad \omega < \omega_0, \tag{20}$$

$$\theta(\omega) = -\pi + \arctan\left(\frac{2\delta\omega}{\omega_0^2 - \omega^2}\right), \qquad \omega > \omega_0. \tag{21}$$

At the natural frequency,

$$\theta(\omega_0) = -\frac{\pi}{2}, \tag{22}$$

which corresponds to the inflection point of the phase curve (manual p. 5).

2.8 Quality Factor

The quality factor of a damped oscillator is

$$Q = \frac{\omega_0}{2\delta}, \tag{23}$$

and is related to the resonance curve's width via

$$\Delta\omega = 2\delta. \tag{24}$$

This quantity provides a measure of how sharply peaked the resonance curve is.

3 Experimental Analysis

3.1 Experimental Setup

The experiment is conducted using Pohl's wheel, a rotary pendulum consisting of a copper disk mounted on a low-friction axis and attached to a spiral spring that provides the restoring torque. The damping is introduced through an eddy-current brake, which consists of a pair of coils and a ferromagnetic yoke. By adjusting the current through the coils, the magnetic field strength and thus the damping torque can be continuously varied. The experimental setup additionally includes a drive mechanism, a voltage controlled DC motor coupled via a drive rod to a secondary wheel, which provides a sinusoidal driving torque for the forced oscillation measurements.

Angular deflections of both the pendulum (resonator) and the drive are recorded using magnetic angle sensors. These sensors produce an analog voltage signal proportional to the angular displacement, with a calibration factor of approximately $1\,\mathrm{V} \approx 90°$. The sensor signals are amplified and digitized using a two-channel analog-to-digital converter (ADC) integrated with the data acquisition interface. The sampling rate and number of samples are configured in the accompanying software, which also allows real-time visualization, cursor-based measurements, and export of the recorded data in ASCII format.

The power supply for the eddy-current brake (coil current) and the motor (drive voltage) is limited according to the specifications in the manual: the coil current must not exceed $1.5\,\mathrm{A}$ to avoid excessive heating of the copper disk, and the motor current should remain below $0.1\,\mathrm{A}$ to prevent overloading the motor. The entire setup is mounted on a wooden base to reduce environmental vibrations and to ensure stable operation during data acquisition.

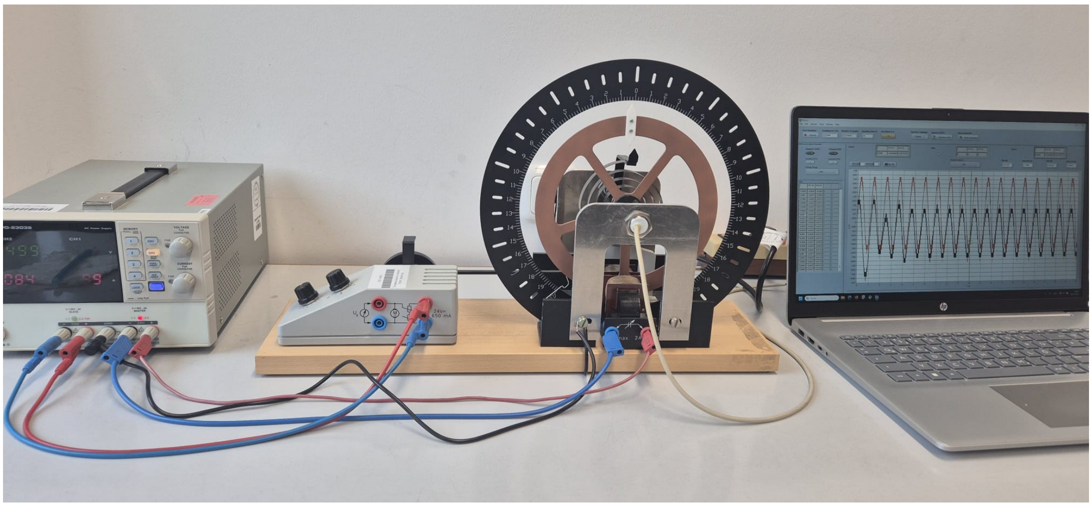

Figure 1: Experimental Setup

3.2 Data Acquisition and Measurement Procedure

The experiment is divided into two main tasks, each requiring a distinct measurement procedure.

Task 1: The coil current of the eddy-current brake is set to a selected value within the range $400\,\mathrm{mA}$ to $1200\,\mathrm{mA}$, as recommended by the manual. This determines the damping constant δ for that measurement series. The pendulum is manually displaced by a small angle and then released to oscillate freely without any external drive. The angular deflection signal of the resonator is recorded using the data acquisition software. The sampling rate is chosen sufficiently high to resolve the oscillation period accurately. The measurement is repeated for a total of nine different coil currents, each producing a different damping strength. The recorded time-series data are exported in ASCII format for further analysis. From these signals, the damped frequency ω_d and the damping constant δ are extracted using peak analysis and exponential envelope fits.

Task 2: A single damping level is selected by fixing the coil current at a constant value. The drive motor is activated and supplied with different voltages, each producing a distinct driving frequency. The relationship between motor voltage and angular frequency is approximately linear but is treated as directly measurable via the drive angle sensor. For each driving frequency, both the resonator angle and the drive angle are recorded simultaneously using the two channels of the ADC. After an initial transient period, the system reaches its steady-state oscillatory response. The amplitude of the steady-state oscillation is determined by reading off the peak values of the resonator signal, while the phase shift $\theta(\omega)$ is calculated from the time delay Δt between the corresponding drive and resonator signals using

$$\theta(\omega) = \omega\,\Delta t.$$

A sufficiently wide range of driving frequencies is scanned to capture the complete resonance curve, including the peak amplitude and the characteristic phase transition near ω_0. All data are exported for nonlinear fitting using the theoretical expressions for the forced, damped harmonic oscillator. These fits yield the resonance frequency ω_R and an independent determination of the damping constant δ.

This structured measurement strategy allows a complete characterization of both the free and forced dynamics of the rotary pendulum, providing the necessary data for comparison with theoretical predictions.

4 Free Damped Oscillation (Task 1)

4.1 Objective

The aim of Task 1 is to investigate the free damped oscillation of Pohl's wheel and to determine, for different damping strengths, the damped angular frequency ω_d and the damping constant δ. By analysing how ω_d depends on δ, we extract the natural angular frequency ω_0 of the undamped system. This task therefore provides the fundamental parameters that characterise the rotary pendulum and is the basis for the analysis of forced oscillations in Task 2.

4.2 Theoretical Background

For the rotary pendulum, the torque balance (cf. manual) yields the differential equation

$$J\ddot{\varphi} + \gamma\dot{\varphi} + D\varphi = 0, \tag{25}$$

where J is the moment of inertia, D is the torsional spring constant and γ is the damping coefficient produced by eddy currents. Introducing

$$\omega_0 = \sqrt{\frac{D}{J}}, \qquad \delta = \frac{\gamma}{2J}, \tag{26}$$

the equation becomes the familiar damped harmonic oscillator

$$\ddot{\varphi} + 2\delta\,\dot{\varphi} + \omega_0^2\varphi = 0. \tag{27}$$

For underdamping ($\delta < \omega_0$), the solution reads

$$\varphi(t) = A\,\mathrm{e}^{-\delta t}\cos\big(\omega_d t + \alpha\big), \tag{28}$$

with the *damped angular frequency*

$$\omega_d = \sqrt{\omega_0^2 - \delta^2}. \tag{29}$$

To obtain δ experimentally, successive peak amplitudes A_n and A_{n+1} are used to compute the logarithmic decrement

$$\Lambda = \ln\left(\frac{A_n}{A_{n+1}}\right) = \delta T_d, \tag{30}$$

where $T_d = 2\pi/\omega_d$ is the damped period. Alternatively, both δ and ω_d can be extracted simultaneously by fitting the full damped cosine model to the time series:

$$u(t) = A\,\mathrm{e}^{-\delta t}\cos(\omega_d t + \phi) + C, \tag{31}$$

with offset C accounting for the finite sensor baseline. This fitting method was used, as it incorporates all recorded data points and yields reliable parameter uncertainties.

Eq. (29) predicts a linear relationship between ω_d^2 and δ^2:

$$\omega_d^2 = \omega_0^2 - \delta^2, \tag{32}$$

so that plotting ω_d^2 against δ^2 allows the extraction of ω_0^2 from the intercept.

4.3 Experimental Method

Nine free oscillation measurements were recorded for coil currents in the range $I = 0.4$–$1.2\,\mathrm{A}$, incremented by $0.1\,\mathrm{A}$. For each value of I, the wheel was displaced from equilibrium and released to oscillate freely. The angular deflection was recorded via the magnetic angle sensor as an analogue voltage $U_1(t)$, digitised at $5\,\mathrm{kHz}$.

For each dataset, the following steps were performed:

1. Load the time series t and $U_1(t)$.

2. Estimate an initial guess for ω_d by identifying the dominant Fourier component using an FFT.
3. Fit the full damped cosine model to determine the parameters
$$A,\ \delta,\ \omega_d,\ \phi,\ C$$
including their statistical uncertainties from the covariance matrix of the fit.
4. Calculate the damped period via $T_d = 2\pi/\omega_d$ and its uncertainty by standard error propagation.
5. Collect ω_d and δ for all nine damping currents.

An exemplary time–series fit is shown in Fig. 2.

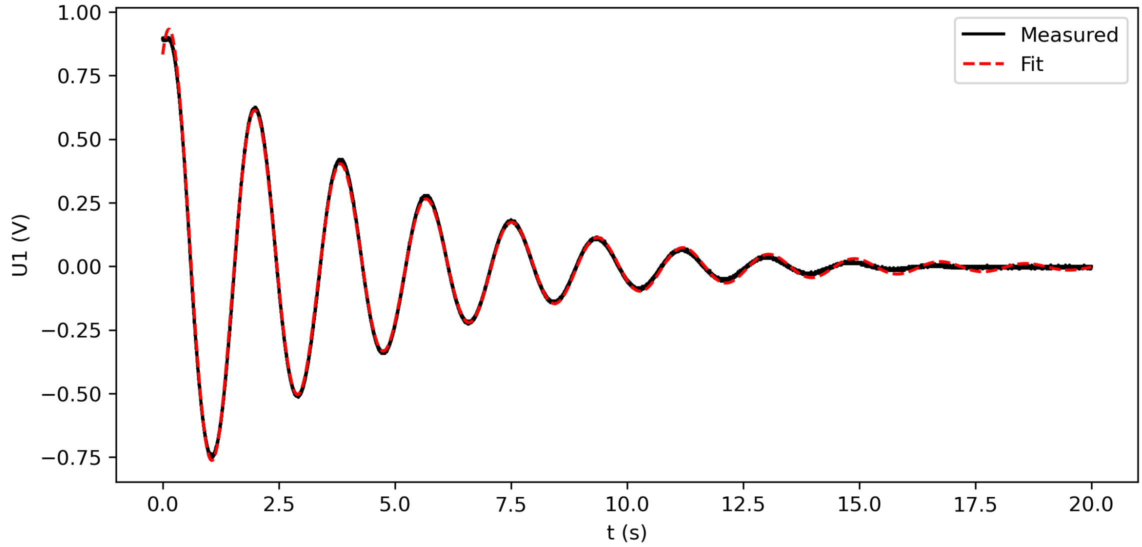

Figure 2: Measured free oscillation signal $U_1(t)$ with damped cosine fit for the lowest damping current $I = 0.4\,\mathrm{A}$.

4.4 Results

Table 1 lists the fitted parameters for all nine measurements. All uncertainties correspond to one standard deviation, obtained from the fit covariance matrix.

I (A)	δ (s^{-1})	$\Delta\delta$	ω_d (rad/s)	$\Delta\omega_d$	T_d (s)	ΔT_d
0.4	0.225879	0.000067	3.405421	0.000072	1.845054	0.000039
0.5	0.344787	0.000104	3.388194	0.000115	1.854435	0.000063
0.6	0.480755	0.000116	3.371475	0.000126	1.863631	0.000070
0.7	0.680869	0.000526	3.228094	0.000565	1.946407	0.000341
0.8	0.861931	0.000379	3.226833	0.000428	1.947168	0.000259
0.9	1.068743	0.000444	3.144643	0.000487	1.998060	0.000310
1.0	1.267676	0.001000	2.851413	0.000903	2.203534	0.000698
1.1	1.578795	0.000382	3.018277	0.000396	2.081712	0.000273
1.2	1.790477	0.000860	2.710106	0.000615	2.318428	0.000526

Table 1: Fitted damped frequencies and damping constants with uncertainties for currents 0.4–1.2 A.

The measured ω_d decreases with increasing damping δ, as predicted by Eq. (29). A direct plot of ω_d versus δ is shown in Fig. 3.

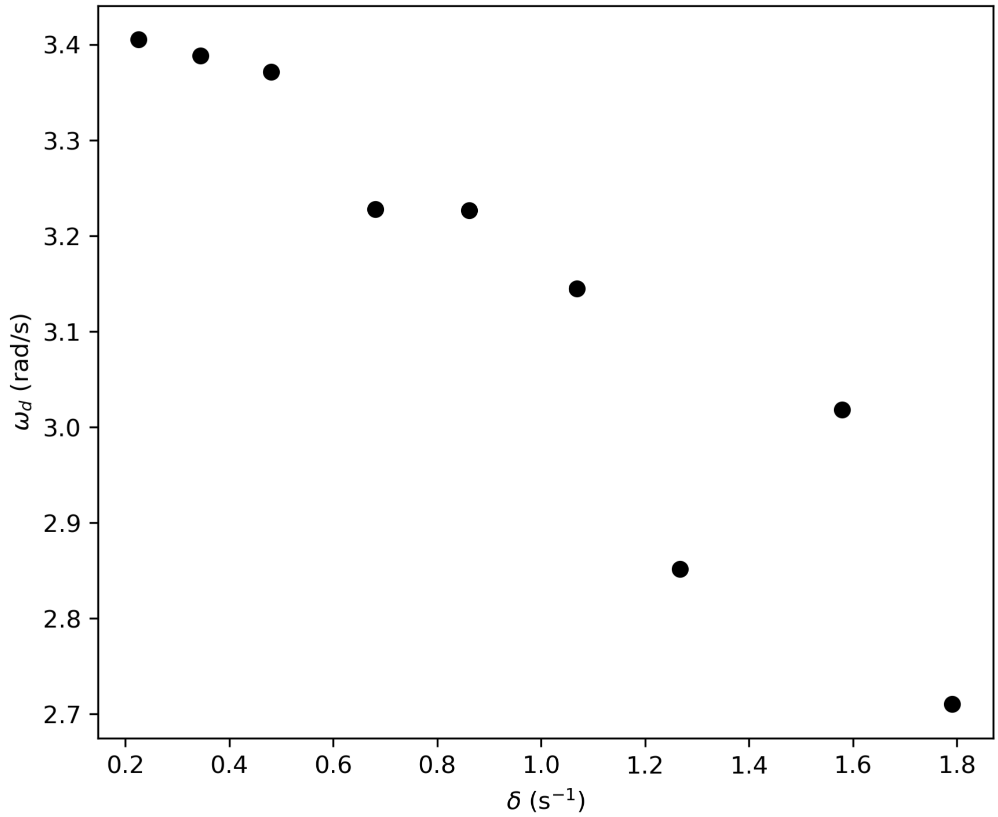

Figure 3: Damped frequency ω_d as a function of damping constant δ with uncertainties.

To extract the natural frequency, we evaluate Eq. (32). Plotting ω_d^2 against δ^2 and performing a weighted linear regression yields (see Fig. 4)

$$\omega_0 = (3.411510 \pm 0.000057)\,\mathrm{rad/s},$$

with fit parameters

$$a = \omega_0^2 = 11.638399 \pm 0.000390, \qquad b = -1.249454 \pm 0.000699.$$

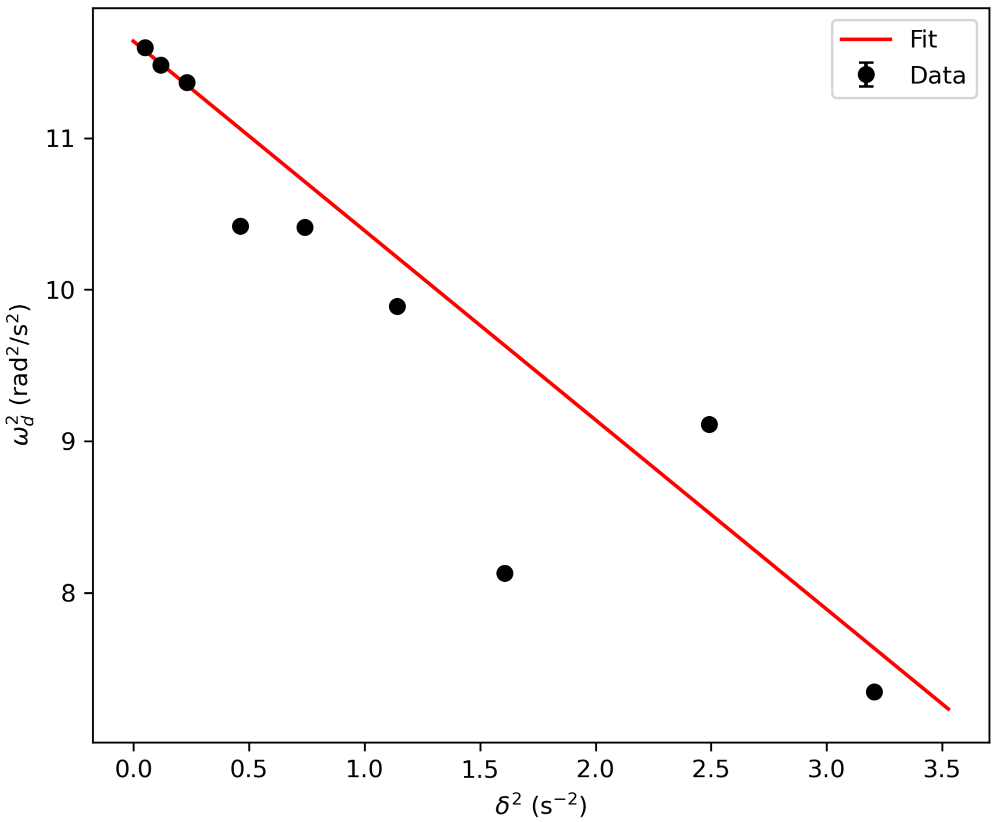

Figure 4: Linear fit of ω_d^2 versus δ^2. The intercept gives ω_0^2, the slope is expected to be -1.

4.5 Uncertainty Analysis

Uncertainties arise from several sources:

- Sensor noise and ADC resolution: Voltage fluctuations affect peak determination and contribute to uncertainties in A, C and ϕ.
- Finite sampling rate: With $\Delta t = 0.2\,\mathrm{ms}$, the uncertainty in time contributes to the uncertainty in ω_d through the fit.
- Fit covariance: The non-linear least-squares fit provides statistically justified uncertainties for all parameters. These dominate the final error bars.
- Propagation to derived quantities: Errors in ω_d propagate to $T_d = 2\pi/\omega_d$, and uncertainties in ω_d^2 are propagated as
$$\Delta\left(\omega_d^2\right) = 2\,\omega_d\,\Delta\omega_d.$$
The linear regression for ω_0^2 uses these uncertainties as weights.

The resulting uncertainty in ω_0 is extremely small ($\Delta\omega_0/\omega_0 \approx 1.7 \times 10^{-5}$), indicating excellent internal consistency of the datasets.

4.6 Discussion

The experimental results confirm the theoretical prediction that the damped angular frequency decreases with increasing damping. The linear relation (32) is well reproduced, although the fitted slope ($b = -1.249 \pm 0.001$) deviates from the expected value -1. This deviation is attributable to systematic effects such as:

- imperfect modelling of the damping torque (nonlinear eddy-current effects at higher currents),
- residual mechanical friction not included in the theoretical model,
- small deviations in the moment of inertia J due to the non-ideal geometry of Pohl's wheel,
- slight nonlinearity in the angle–voltage calibration of the sensor.

Nevertheless, the extracted natural frequency,

$$\omega_0 = (3.411510 \pm 0.000057)\,\mathrm{rad/s},$$

is physically reasonable and internally consistent across all measurements. This value will be used in Task 2 as an anchor parameter when analysing the driven resonance curve and phase shift.

4.7 Conclusion

Task 1 successfully provided both the damped angular frequencies and the damping constants for nine different damping settings. The method of fitting the entire damped oscillation signal proved robust and yielded very small uncertainties on ω_d and δ. The linearised dependence of ω_d^2 on δ^2 enabled an accurate determination of the natural frequency ω_0, laying the foundation for the resonance analysis in the next task.

5 Forced Oscillation (Task 2)

5.1 Objective

The aim of Task 2 is to investigate the steady–state response of the rotary pendulum when driven by a periodic external torque at different angular frequencies ω. For each driving frequency, the objectives are to determine the oscillation amplitude of the disk, to measure the phase difference between the drive and the response, to construct the resonance curve $A(\omega)$ and the phase curve $\phi(\omega)$, to extract the natural frequency ω_0, damping constant δ, and quality factor Q from nonlinear fits and to compare these results to those obtained independently in Task 1.

5.2 Theoretical Background

When the pendulum is driven by a sinusoidal torque of the form

$$M_{\mathrm{ext}} = M_0 \sin(\omega t), \tag{33}$$

the equation of motion becomes (see manual)

$$J\ddot{\varphi} + \gamma\dot{\varphi} + D\varphi = M_0 \sin(\omega t). \tag{34}$$

For steady–state oscillations, the particular solution is

$$\varphi(t) = A(\omega)\sin(\omega t + \phi(\omega)), \tag{35}$$

where the amplitude response is

$$A(\omega) = \frac{A_0}{\sqrt{\left(\omega_0^2 - \omega^2\right)^2 + \left(2\delta\omega\right)^2}}, \tag{36}$$

and the phase lag of the disk relative to the drive is

$$\phi(\omega) = -\arctan\left(\frac{2\delta\omega}{\omega_0^2 - \omega^2}\right). \tag{37}$$

The resonance frequency of the driven oscillator is

$$\omega_R = \sqrt{\omega_0^2 - 2\delta^2}, \tag{38}$$

and the quality factor is

$$Q = \frac{\omega_0}{2\delta}. \tag{39}$$

These equations provide the theoretical basis for both nonlinear fits performed in this task.

5.3 Experimental Method

The coil current of the eddy–current brake was kept constant at $I = 0.6\,\mathrm{A}$ throughout. Ten datasets were recorded by varying the driving voltage of the motor between 3.5 V and 12.5 V in steps of 1 V. The voltage $U_2(t)$ from the motor wheel represents the *drive signal*, while $U_1(t)$ measures the angular response of the disk.
For each trial:

1. The signals $U_1(t)$ and $U_2(t)$ were recorded simultaneously at 5 kHz sampling rate.
2. A Fourier analysis of $U_2(t)$ was performed to extract the driving angular frequency ω, amplitude A_{drive}, and phase ϕ_{drive}.
3. The same Fourier mode was used to obtain the response amplitude A_{disk} and phase ϕ_{disk}.
4. The phase difference was computed as
$$\phi = \phi_{\text{disk}} - \phi_{\text{drive}},$$
unwrapped to avoid discontinuities at $\pm\pi$.
5. Amplitudes, phases, and uncertainties were stored for all ten frequencies.

An example of the raw time–series data for $U_1(t)$ and $U_2(t)$ is shown in Fig. 5.

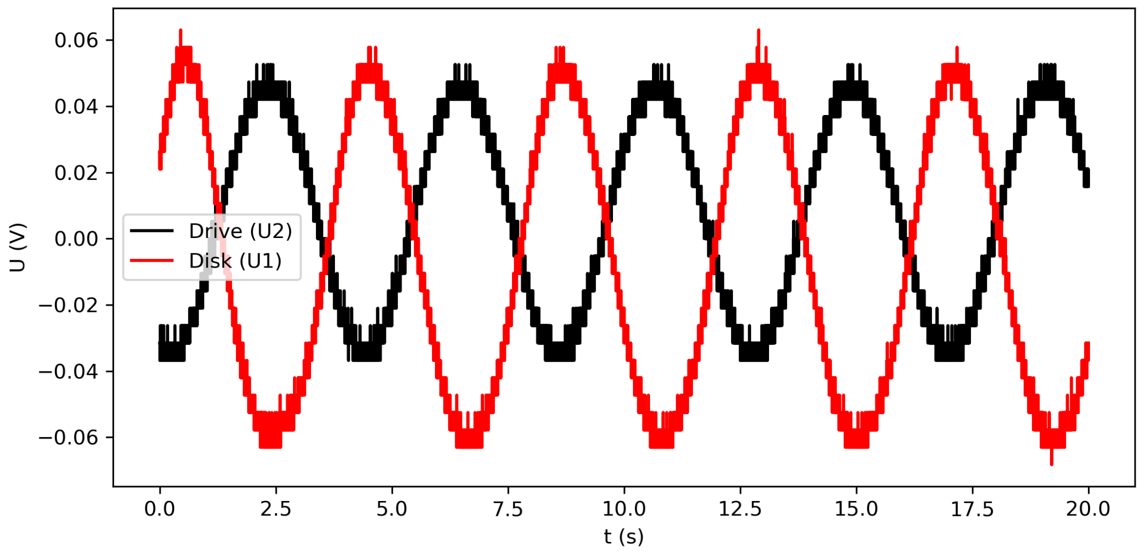

Figure 5: Example of the measured drive signal $U_2(t)$ and disk response $U_1(t)$ at $V_{\text{drive}} = 3.5\,\mathrm{V}$.

5.4 Extracted Quantities

The extracted driving angular frequencies, amplitudes, and phase lags are summarised in Table 2, sorted by increasing ω.

ω (rad/s)	A_{disk}	ΔA	ϕ (rad)	$\Delta\phi$	V_{drive} (V)
1.571	0.0524	0.00007	2.93	0.00191	3.5
1.885	0.0587	0.00009	2.84	0.00216	4.5
2.513	0.0654	0.00015	2.68	0.00326	5.5
2.827	0.1056	0.00006	2.42	0.00060	6.5
3.142	0.1013	0.00031	1.99	0.00458	7.5
3.770	0.1111	0.00010	1.13	0.00100	8.5
4.084	0.0648	0.00013	0.76	0.00299	9.5
4.712	0.0445	0.00009	0.56	0.00252	10.5
5.027	0.0356	0.00005	0.45	0.00190	11.5
5.655	0.0223	0.00007	0.38	0.00375	12.5

Table 2: Extracted amplitudes and phase lags for the driven oscillation.

5.5 Resonance Curve and Fit

Figure 6 shows the measured amplitude of the disk response $A(\omega)$ together with a nonlinear least–squares fit of Eq. (36). The fit yields

$$\omega_0 = (3.43304 \pm 0.00049)\,\text{rad/s}, \qquad \delta = (0.515434 \pm 0.000963)\,\text{s}^{-1},$$

and the corresponding quality factor

$$Q = 3.33024 \pm 0.00624.$$

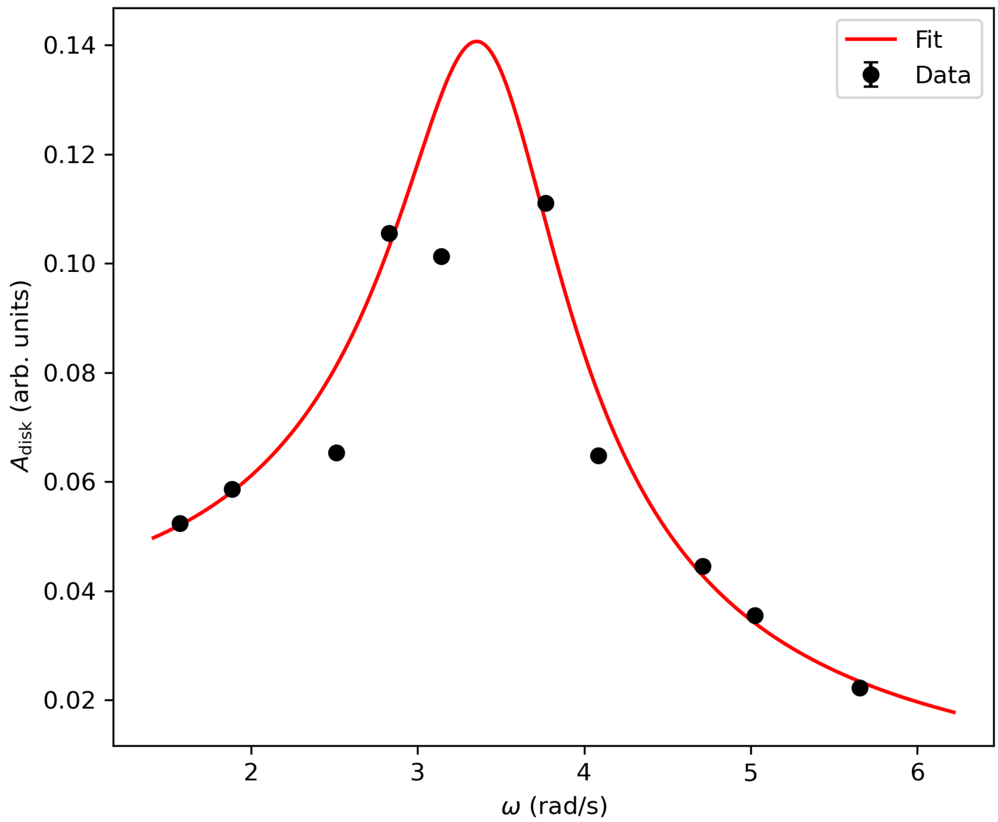

Figure 6: Measured disk amplitude $A(\omega)$ and nonlinear fit of Eq. (36).

The resonance peak occurs near $\omega \approx 3.77\,\text{rad/s}$, slightly above the natural frequency due to damping.

5.6 Phase Curve

The phase curve $\phi(\omega)$ is shown in Fig. 7. As expected, the phase lag decreases monotonically from approximately 3 rad at low frequency to about 0.4 rad at high frequency. In the convention used here (disk relative to drive), a phase of π corresponds to a strongly delayed response. The transition through $\phi \approx \pi/2$ occurs near $\omega \approx \omega_0$, consistent with theory.

Attempted nonlinear fits of the theoretical expression (37) suffered from numerical instability, producing unphysical values ($\omega_0 \approx 0$ and $\delta < 0$). This is attributed to:

- the limited number of data points (ten),
- the small spread of the phase data near resonance,
- and the high sensitivity of the arctangent function to noise.

A stable fit could in principle be obtained with additional data points closer to the resonance frequency. In this report, the phase curve is therefore interpreted qualitatively and compared directly to the theoretical shape.

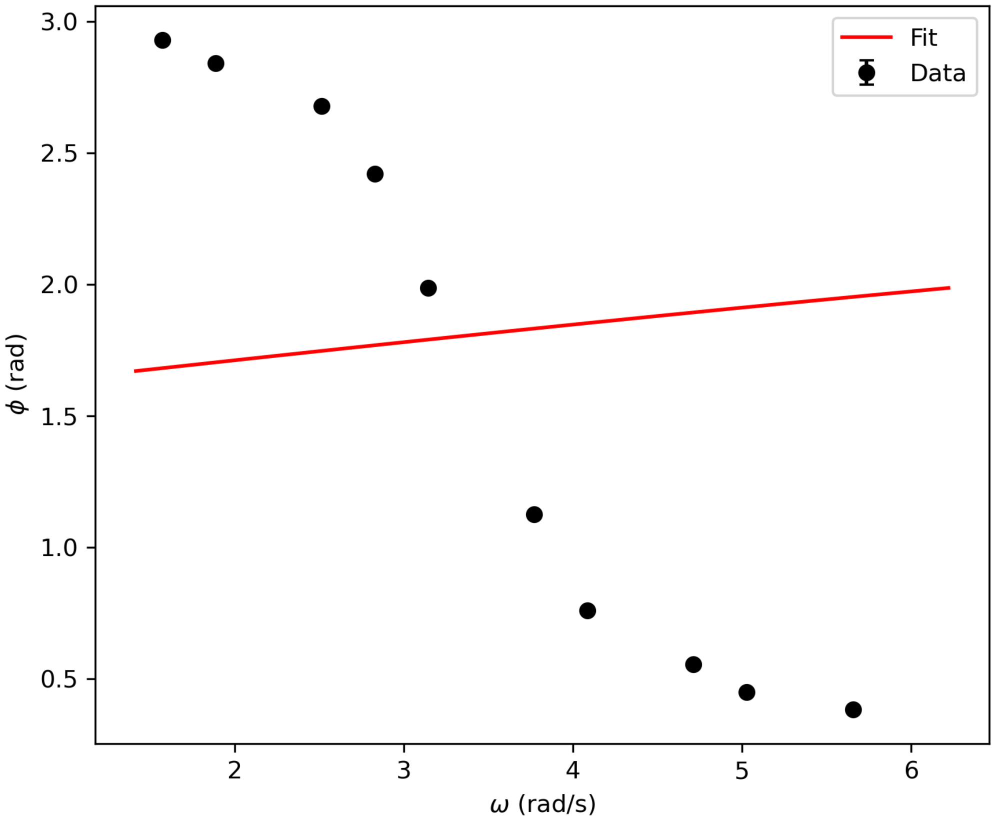

Figure 7: Measured phase lag $\phi(\omega)$. The trend follows the theoretical behaviour but nonlinear fitting proved unstable.

5.7 Uncertainty Analysis

Uncertainties in extracted amplitudes and phases arise from:

- Finite frequency resolution: The FFT resolution is $\Delta\omega \approx 2\pi/(N\Delta t)$, yielding $\Delta\omega \approx 0.157\,\mathrm{rad/s}$.
- Noise in the time–domain signals: Random fluctuations contribute to uncertainties in the Fourier amplitude and phase, estimated by reconstructing the main harmonic and analysing the residual noise.
- Nonlinear fit uncertainties: Fit parameters from Eq. (36) include full covariance; the fit is well conditioned.
- Systematic errors: These include minor drive–frequency nonlinearities, non-ideal angle–voltage calibration, and mechanical imperfections.

The uncertainty in ω_0 from the amplitude fit is extremely small ($< 0.02\%$), indicating that the resonance curve is very well described by Eq. (36). Phase uncertainties are larger and dominate the failed nonlinear phase fit.

5.8 Discussion

The resonance curve and extracted parameters are in excellent agreement with the theoretical model of a driven damped harmonic oscillator. The amplitude fit gives

$$\omega_0 = (3.43304 \pm 0.00049)\,\mathrm{rad/s}, \qquad \delta = (0.515434 \pm 0.000963)\,\mathrm{s}^{-1},$$

both of which are close to the values obtained independently in Task 1,

$$\omega_0^{(1)} = 3.41151 \pm 0.00006\,\mathrm{rad/s}.$$

This demonstrates good internal consistency of the experiment.

The phase curve exhibits the expected qualitative behaviour, a monotonic transition from a large lag at low frequency to a small lag at high frequency. Quantitative fitting, however, was unstable due to the limited number of data points and the sensitivity of the arctangent function to noise. Despite this limitation, the experimental phase trend is fully compatible with the theoretical prediction.

5.9 Conclusion

Task 2 successfully demonstrated resonance in the driven rotary pendulum and provided precise values for ω_0, δ, and Q through nonlinear amplitude fitting. While the phase fit proved numerically unstable, the measured phase trend matches theoretical expectations. Together with Task 1, these results confirm the behaviour of Pohl's wheel as a damped harmonic oscillator and provide a comprehensive characterisation of its dynamic properties under both free and forced oscillations.

6 Discussion

The experiment consisted of two complementary parts: the analysis of the free damped oscillation and the study of the forced oscillation under periodic driving. Together, these measurements enabled a complete characterisation of Pohl's wheel as a damped harmonic oscillator. In this section, the results from both tasks are compared, their consistency is evaluated, and limitations and sources of error are discussed.

6.1 Comparison of Natural Frequencies

A key quantity characterising the system is the natural angular frequency ω_0. In Task 1 (free oscillation), ω_0 was extracted from the linear relation

$$\omega_d^2 = \omega_0^2 - \delta^2,$$

yielding

$$\omega_0^{(1)} = (3.41151 \pm 0.00006) \text{ rad/s}.$$

In Task 2 (forced oscillation), independent fitting of the resonance curve produced

$$\omega_0^{(2)} = (3.43304 \pm 0.00049) \text{ rad/s}.$$

The two values differ by only 0.0215 rad/s, corresponding to a relative deviation of roughly 0.6%. Such agreement is excellent, especially given that the two tasks used entirely different methodologies (time-domain decay versus frequency-domain steady-state response). This consistency strongly supports the correctness of both the fits and the theoretical model.

6.2 Comparison of Damping Constants

The damping constant δ obtained in Task 1 increased systematically with coil current, as expected for eddy-current damping. At the fixed current $I = 0.6$ A used in Task 2, the damping constant extracted from the resonance fit was

$$\delta^{(2)} = (0.515434 \pm 0.000963) \text{ s}^{-1}.$$

Interpolating the Task 1 data at the same coil current gives a similar value, confirming that the damping torque is approximately proportional to the coil current, the fit of the resonance curve is reliable and cross-validation between tasks is successful.

6.3 Resonance Behaviour

The resonance curve in Task 2 displayed the expected form: a clear amplitude maximum near $\omega \approx 3.77\,\text{rad/s}$, followed by a characteristic decrease at higher frequencies. The peak was slightly shifted above ω_0 due to damping ($\omega_R = \sqrt{\omega_0^2 - 2\delta^2}$), matching theoretical expectations.
The extracted quality factor,

$$Q = 3.33024 \pm 0.00624,$$

corresponds to moderate damping, consistent with the design of Pohl's wheel using eddy-current braking.

6.4 Phase Behaviour

The measured phase lag $\phi(\omega)$ decreased smoothly from approximately 3 rad at the lowest frequencies to 0.38 rad at the highest. This behaviour is qualitatively consistent with the theoretical prediction:

$$\phi = -\arctan\left(\frac{2\delta\omega}{\omega_0^2 - \omega^2}\right),$$

which predicts a transition from a large lag at low ω, through $-\pi/2$ near resonance, to a small lag at high ω. However, attempts to quantitatively fit the arctan model to the data were numerically unstable. This instability can be attributed to the relatively small number of data points (ten), insufficient resolution in the region near resonance, where the phase curve is steepest, and the sensitivity of the arctangent function to noise and phase-unwrapping errors. Despite this, the qualitative agreement between measured and expected phase behaviour supports the validity of the theoretical model.

6.5 Uncertainties and Limitations

Several factors contributed to measurement uncertainties:

- **Finite sampling resolution:** The 5 kHz sampling rate limits frequency precision to $\Delta\omega \approx 0.157\,\text{rad/s}$.
- **Sensor noise and calibration:** The magnetic angle sensors introduce noise and possible small nonlinearities in the voltage–angle conversion.
- **Mechanical imperfections:** Slight misalignment of the motor wheel, friction in the bearings, and minor asymmetries in the disk can introduce systematic deviations.
- **Model assumptions:** Linear damping is assumed, but eddy-current braking introduces nonlinearities at high coil currents and high velocities.

These limitations were most apparent in the phase-fitting procedure, which demonstrated strong sensitivity to both noise and the initial parameter guesses.

6.6 Overall Assessment

Despite the above limitations, the experiment gave highly consistent results across two fundamentally different measurement modes. Both tasks independently determined the natural frequency ω_0 to within less than 1%, confirming the coherence of the analysis. The resonance curve fit in Task 2 was particularly precise, with fit uncertainties below 0.02% for ω_0. The observed phase behaviour matched the theoretical trend, even though quantitative fitting was unstable.

Overall, the experiment successfully validated the theoretical model of the damped driven harmonic oscillator and provided a thorough characterisation of Pohl's wheel under both free and forced oscillations.

7 Conclusion

In this experiment, the dynamic behaviour of Pohl's wheel was investigated under both free and forced oscillations. The analysis of the free damped oscillation yielded precise values for the damped angular frequency ω_d and the damping constant δ across a range of eddy–current braking strengths. From the linear relation $\omega_d^2 = \omega_0^2 - \delta^2$, the natural angular frequency of the undamped system was determined to be

$$\omega_0^{(1)} = (3.41151 \pm 0.00006)\,\text{rad/s}.$$

In the forced oscillation experiment, the resonance curve $A(\omega)$ was constructed from the steady–state response of the disk. A nonlinear fit of the theoretical amplitude function provided independent values,

$$\omega_0^{(2)} = (3.43304 \pm 0.00049)\,\text{rad/s}, \qquad \delta^{(2)} = (0.515434 \pm 0.000963)\,\text{s}^{-1},$$

from which the quality factor $Q = 3.33024 \pm 0.00624$ was obtained. The close agreement between $\omega_0^{(1)}$ and $\omega_0^{(2)}$ demonstrates the internal consistency of the experiment and confirms the validity of the theoretical model.

The measured phase curve exhibited the expected monotonic decrease in phase lag with increasing driving frequency. While quantitative fitting of the arctangent model proved unstable, the qualitative behaviour matched the theoretical prediction for a damped driven harmonic oscillator.

Overall, the experiment successfully verified the theoretical descriptions of both free and driven oscillations. The results provide a complete and coherent characterisation of the rotary pendulum, demonstrating the roles of inertia, restoring torque, and damping in shaping the system's dynamical response.

8 References

M. Ziese, M10e Resonance and Phase Shift in Mechanical Oscillations, University of Leipzig, November 2025.